Unit-1
Introduction to Research Methodology

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Meaning of Research

- Word 'Research' is comprises of two words **= Re+Search (**It means to search again.)
- So, research means a systematic investigation or activity to gain new knowledge of the already existing facts.
- Research in common parlance refers to a search for knowledge. Once can also define research as a scientific and systematic search for pertinent information on a specific topic
- Research is an intellectual activity. It is responsible for bringing to light new knowledge
  - It is also responsible for correcting the present mistakes, removing existing misconceptions and adding new learning to the existing fund of knowledge

Meaning of Research

- Research is also considered as the application of scientific method in solving the problems.
- It is a systematic, formal and intensive process of carrying on the scientific method of analysis.
- There are many ways of obtaining knowledge. They are intuition, revelation, and authority, logical manipulation of basic assumptions, informed guesses, observation, and reasoning by analogy.
- One of the branches of research known as empirical research is highly goal-oriented technique.

Definitions of Research: By Different Author

- Research is an endeavor / attempt to discover, develop and verify knowledge. It is an intellectual process that has developed over hundreds of years ever changing in purpose and form and always researching to truth **- J. Francis Rummel**
- Research is an honest, exhaustive, intelligent searching for facts and their meanings or implications with reference to a given problem. The product or findings of a given piece of research should be an authentic, verifiable contribution to knowledge in the field studied **-P.M. Cook**

Definitions of Research: By Different Author

- Research may be defined as a method of studying problems whose solutions are to be derived partly or wholly from facts. - **W.S. Monroes**
- Research is considered to be the more formal, systematic intensive process of carrying on the scientific method of analysis. It involves a more systematic structure of investigation, usually resulting in some sort of formal record of procedures and a report of results or conclusion**.- John W. Best**

Definitions of Research: By Different Author

- Research comprises defining and redefining problems ,formulating hypothesis or suggested solutions, collecting ,organizing and evaluating data, making deductions and reaching conclusions and at last careful testing the conclusions to determine whether they fit the formulated hypothesis.- **Clifford Woody**
- Research is a systematic effort to gain new knowledge.- **Redman & Mori**

Definitions of Research: By Different Author

- Social research may be defined as a scientific undertaking which by means of logical and systematized techniques aims to discover new facts or verify and test old facts , analyze their sequences , interrelationships and casual explanation which were derived within an appropriate theoretical frame of reference , develop new scientific tools , concepts and theories which would facilitate reliable and valid study of human behavior **- P.V. Younge**
- According to Clifford Woody research comprises defining and redefining problems, formulating hypothesis or suggested solutions; collecting, organizing and evaluating data; making deductions and reaching conclusions; and at last, carefully testing the conclusions to determine whether they fit the formulating hypothesis

Objective / Purpose of Research

- To gain familiarity with a phenomenon or to achieve new insights into it. (Studies with this object in view are termed as exploratory or formative research studies).
- To portray accurately the characteristics of a particular individual, situation or a group.(Studies with this object in view are known as descriptive research studies).

Objective/ Purpose of Research

- To determine the frequency with which something occurs or with which it is associated with something else. (Studies with this object in view are known as diagnostic research studies)
- To test a hypothesis of a causal relationship between variables. (Such studies are known as hypothesis-testing research studies).

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Motivation in Research

What makes people to undertake research?

- Desire to get a research degree along with its consequential benefits
- Desire to face the challenge in solving the unsolved problems, i.e., concern over practical problems initiate's research
- Desire to get intellectual joy of doing some creative work
- Desire to be of service to society
- Desire to get respectability.

Many more factors such as **directives of government**, **employment conditions**, **curiosity about new things**, **desire to understand causal relationships**, **social thinking** and **awakening**, and the like may as well **motivate** (or at times compel) people to perform research operations.

Characteristics of Research:

- Research is directed toward the solution of a problem.
- Research requires expertise.
- Research emphasizes the development of generalizations, principles, or theories that will be helpful in predicting future occurrences.
- Research is based upon observable experience or empirical evidences.
- Research demands accurate observation and description

Characteristics of Research:

- Research involves gathering new data from primary or first-hand sources or using existing data for a new purpose.
- Research is characterized by carefully designed procedures that apply rigorous analysis.
- Research involves the quest for answers to un-solved problems.
- Research strives to be objective and logical, applying every possible test to validate the procedures employed the data collected and the conclusions reached.
- Research is characterized by patient and unhurried activity.
- Research is carefully recorded and collected.
- Research sometimes requires courage.

Types of Research:

- Descriptive Research Vs. Analytical Research
- Applied Research vs Fundamental Research
- Quantitative Research vs. Qualitative Research
- Conceptual Research vs. Empirical Research
- Longitudinal Research
- Cross-Sectional Research
- Philosophical Research
- Historical Research
- Survey Research
- Experimental Research
- Case-Study Research
- clinical or diagnostic research

Types of Research: Descriptive Research

Descriptive research focuses on documenting and describing the current state of a phenomenon or population. It answers the question, “What is happening?” rather than “Why is it happening?”

Characteristics:

- Provides a snapshot of reality by detailing features, behaviors, or characteristics.
- Does not investigate causes or relationships.
- Collects data that reflects the present status without manipulation or control.
- Commonly used methods include surveys, case studies, and observations.
- Results are often presented as statistics, frequencies, or descriptive summaries.

Types of Research: Descriptive Research

When to Use Descriptive Research:

- You want to map out or profile a situation, event, or population.
- Your goal is to establish baseline information.
- You are exploring phenomena for the first time and need to define what's happening.
- You need straightforward, factual reporting without interpreting causes.

Examples:

- Reporting demographic characteristics of a community.
- Documenting student attendance and grades in a semester.
- Surveying consumer preferences in a market.

Types of Research: Analytical Research

Analytical research goes beyond description to examine relationships, causes, and effects. It focuses on answering the question "Why or how does this happen?"

Characteristics:

- Uses existing data or newly collected information to test hypotheses or theories.
- Employs methods such as statistical analysis, experiments, and longitudinal studies.
- Focuses on interpreting data to draw conclusions about underlying patterns and causal links.
- Requires a deeper level of analysis to explain findings.

Types of Research: Analytical Research

When to Use Analytical Research:

- You want to understand reasons behind observed patterns or behaviors.
- You need to test theories or explore cause-and-effect relationships.
- Your research question involves predicting outcomes or explaining phenomena.
- You aim to provide insights that support decision-making or further research.

Examples:

- Investigating the factors influencing student performance in online learning.
- Analyzing causes of customer satisfaction variations in different regions.
- Examining correlations between health behaviors and disease incidence.

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Types of Research:

Aspect	Descriptive Research	Analytical Research
Purpose	Describe characteristics or current state	Explain reasons, causes, or relationships
Research Question	What is happening?	Why or how is it happening?
Data Collection	Surveys, observations, case studies	Experiments, statistical analysis, secondary data analysis
Data Analysis	Summary statistics, frequencies	Hypothesis testing, correlations, regression, cause-effect analysis
Outcome	Snapshot of facts	Interpretation and explanations
Typical Usage	Baseline info, profiling, fact-finding	Theory testing, causal explanations, predictions

Types of Research: Basic/Fundamental Research

Basic research, also called fundamental or pure research, seeks to advance theoretical knowledge and understanding without immediate concern for practical applications. It is driven by intellectual curiosity and the desire to understand fundamental principles governing natural and social phenomena.

Characteristics of basic research:

- **Theory-focused:** Develops, tests, or refines theoretical frameworks and conceptual models
- **Knowledge-driven:** Motivated by curiosity and desire to understand rather than solve immediate problems
- **Long-term perspective:** Benefits may not be apparent for years, decades, or generations
- **Discipline-building:** Advances understanding within and across academic fields
- **Methodological innovation:** Often develops new research methods and analytical techniques
- **Publication-oriented:** Results typically disseminated through academic journals and conferences

Types of Research: Basic/Fundamental Research

Types of basic research:

- **Descriptive basic research:** Documents and describes phenomena to build foundational knowledge
- **Exploratory basic research:** Investigates new or poorly understood areas
- **Explanatory basic research:** Develops theories to explain observed patterns and relationships
- **Methodological research:** Creates new research tools, techniques, and analytical approaches

Types of Research: Basic/Fundamental Research

Evaluation criteria for basic research:

- **Theoretical contribution:** Advances understanding of fundamental principles
- **Methodological rigor:** Uses appropriate and innovative research methods
- **Scholarly impact:** Influences future research and theoretical development
- **Peer recognition:** Acceptance by academic community through publications and citations
- **Reproducibility:** Findings can be replicated and verified by other researchers

Types of Research: Applied Research

Applied research addresses specific practical problems and seeks to improve existing conditions or develop solutions that can be implemented in real-world settings. It takes theoretical knowledge and uses it to create tangible benefits for individuals, organizations, or society.

Characteristics of applied research:

- **Problem-focused:** Addresses identified needs, challenges, or opportunities
- **Solution-oriented:** Seeks actionable results that can be implemented
- **Stakeholder involvement:** Often involves people or organizations that will use the results
- **Shorter timelines:** Expected to produce results within specific timeframes
- **Context-specific:** Considers practical constraints and implementation factors
- **Outcome measurement:** Evaluates effectiveness and impact of interventions

Types of Research: Applied Research

Types of applied research:

- **Evaluation research:** Assesses effectiveness of programs, policies, or interventions
- **Action research:** Involves practitioners in identifying problems and testing solutions
- **Translational research:** Applies basic research findings to practical contexts
- **Development research:** Creates new products, services, or processes
- **Policy research:** Informs decision-making and policy development

Types of Research: Applied Research

Evaluation criteria for applied research:

- **Practical utility:** Degree to which findings address real-world problems
- **Implementation success:** Effective application of results in target settings
- **Cost-effectiveness:** Efficient use of resources to achieve desired outcomes
- **Stakeholder satisfaction:** Meeting needs and expectations of intended users
- **Broader impact:** Potential for wider application and social benefit
- **Evidence quality:** Methodological rigor appropriate for intended applications

Types of Research: Quantitative Research

- Quantitative research collects and analyzes numerical data to identify patterns, test hypotheses, and establish relationships between variables.
- This approach emphasizes measurement, statistical analysis, and objective findings that can be generalized to larger populations.

Types of Research: Quantitative Research

<u>Core principles of quantitative research:</u>

- Measurement: Variables are operationally defined and measured using standardized instruments
- Sampling: Large, representative samples enable generalization to broader populations
- Standardization: Consistent procedures ensure reliability and replicability
- Statistical analysis: Mathematical techniques identify patterns and test significance
- Objectivity: Researchers minimize bias through controlled procedures and statistical inference

Types of Research: Quantitative Research

When to use quantitative research:

- You want to test specific hypotheses or theories
- You need results that can be generalized to larger populations
- You're measuring relationships between clearly defined variables
- You want to compare groups or track changes over time
- You need to make predictions or forecasts
- Your field values statistical evidence and replication
- You have access to large samples and standardized measures

Types of Research: Quantitative Research

<u>Sample size considerations:</u>

- **Descriptive studies:** Minimum 30 participants for basic statistics
- **Correlational studies:** Generally 30+ participants per variable analyzed
- **Experimental studies:** Power analysis determines adequate sample size
- **Survey research:** Sample size depends on population size and desired precision
- **Complex analyses:** Larger samples needed for multivariate statistics

Types of Research: Quantitative Research

Data collection methods:

- Online surveys: Cost-effective for large samples but may have selection bias
- Telephone interviews: Good response rates but increasingly difficult to conduct
- Face-to-face interviews: High quality data but expensive and time-consuming
- Mail surveys: Reaches diverse populations but low response rates
- Administrative data: Large samples but limited to available variables

Types of Research: Quantitative Research

<u>Advantages</u>:

- Results can be generalized to larger populations through statistical inference
- Findings are replicable when methods are clearly documented
- Statistical significance testing provides confidence in results
- Efficient data collection from large samples using standardized instruments
- Can identify relationships and patterns across many variables simultaneously
- Enables prediction and forecasting based on statistical models
- Widely accepted in many academic and professional fields

Types of Research: Quantitative Research

Limitations:

- May oversimplify complex social and psychological phenomena
- Limited ability to capture context, meaning, and subjective experiences
- Requires large samples to achieve statistical power and generalizability
- Less flexible once data collection begins
- Risk of measurement error if instruments are not well-designed
- May miss unexpected findings that don't fit predetermined categories
- Can be expensive for large-scale studies requiring representative samples

Types of Research: Qualitative Research

- Qualitative research collects and analyzes non-numerical data to understand experiences, meanings, processes, and social phenomena.
- This approach emphasizes interpretation, context, and in-depth understanding of how people make sense of their world

Types of Research: Qualitative Research

Core principles of qualitative research:

- **Naturalistic inquiry:** Studies phenomena in natural settings rather than controlled environments
- **Inductive analysis:** Develops theories and concepts from patterns observed in data
- **Holistic perspective:** Considers context, relationships, and multiple influences
- **Subjective understanding:** Values participants' perspectives and lived experiences
- **Flexible design:** Adapts methods and questions as understanding develops

Types of Research: Qualitative Research

When to use qualitative research:

- You're exploring new, complex, or poorly understood phenomena
- You want to understand personal experiences, meanings, and interpretations
- You need to explain unexpected quantitative findings
- You're developing new theories or conceptual frameworks
- Your research question asks "how" or "why" rather than "how many"
- You want to capture context, process, and change over time
- You need to understand cultural or social influences

Types of Research: Qualitative Research

<u>Sampling in qualitative research:</u>

- **Purposeful sampling:** Select participants who can provide rich information
- **Theoretical sampling:** Choose participants to develop emerging theories
- **Snowball sampling:** Use referrals to reach hard-to-find populations
- **Maximum variation:** Include diverse perspectives and experiences
- **Sample sizes:** Typically 5-50 participants, depending on approach and saturation

Types of Research: Qualitative Research

Data analysis approaches:

- **Thematic analysis:** Identify patterns and themes across data
- **Content analysis:** Systematic categorization of data
- **Discourse analysis:** Examine language use and social construction
- **Narrative analysis:** Focus on story structure and meaning
- **Constant comparative:** Continuously compare data to develop concepts

Types of Research: Qualitative Research

<u>Advantages:</u>

- Provides rich, detailed understanding of complex phenomena
- Captures context, meaning, and subjective experiences
- Flexible and adaptable throughout the research process
- Can uncover unexpected insights and generate new theories
- Gives voice to participants and marginalized perspectives
- Useful for exploring sensitive topics requiring trust and rapport
- Can inform policy and practice through deep understanding

Types of Research: Qualitative Research

Limitations:

- Findings may not be generalizable to other populations or contexts
- Time-intensive data collection, transcription, and analysis
- Potential for researcher bias in data collection and interpretation
- Difficult to replicate exactly due to contextual factors
- May lack statistical power for making broad claims
- Requires significant interpretive skills and theoretical knowledge
- Can be challenging to establish reliability and validity using traditional criteria

Research Method

- Research method refers to the specific techniques, procedures, and tools utilized by researchers to gather, analyze, and interpret data.
- It encompasses the practical aspects of research, involving the selection and application of appropriate methods to answer research questions or test hypotheses.
- Research methods can vary depending on the nature of the study, the discipline, and the data required.
- Examples of research methods include surveys, experiments, interviews, observations, and case studies.

Research Method

Advantages of Research Method

- Provides structure: Research methods provide a structured approach to conducting research, ensuring that data is collected and analyzed systematically.
- Enhances credibility: The use of rigorous research methods enhances the credibility and reliability of research findings.
- Allows replication: Well-documented research methods enable other researchers to replicate the study, contributing to the validation of results.
- Facilitates comparison: By using standardized research methods, researchers can compare and contrast findings across different studies.
- Enables statistical analysis: Research methods such as experiments and surveys provide data that can be subjected to statistical analysis, yielding meaningful insights.

Research Method

Advantages of Research Method

- Offers flexibility: Research methods can be tailored to suit the specific needs of a study, allowing researchers to adapt and modify techniques as necessary.
- Provides a basis for decision-making: Sound research methods provide a basis for informed decision-making in various fields, such as policy development or business planning.
- Supports theory development: Research methods contribute to the development and refinement of theories by providing empirical evidence.
- Expands knowledge: By employing diverse research methods, researchers can expand knowledge in their respective fields and discover new relationships or phenomena.
- Enables interdisciplinary research: Research methods can be applied across disciplines, facilitating interdisciplinary collaborations and the exploration of complex research questions.

Research Method

Disadvantages of Research Method

- Limited scope: Research methods may have limitations in terms of the specific data they can collect or the phenomena they can investigate.
- Potential bias: Researchers may introduce bias unintentionally through the selection or interpretation of research methods.
- Ethical considerations: Certain research methods, such as experiments involving human subjects, may raise ethical concerns that need to be carefully addressed.
- Time-consuming: Some research methods can be time-consuming to implement, especially when dealing with large sample sizes or complex data analysis.
- Costly: Certain research methods, such as large-scale surveys or experiments, may require significant financial resources to conduct.

Research Method

Disadvantages of Research Method

- Lack of generalizability: Findings from research methods that utilize small sample sizes or specific contexts may not be easily generalized to broader populations or situations.
- Limited accessibility: Access to certain research methods, such as specialized equipment or archives, may be restricted, limiting their application.
- Data limitations: Research methods may not capture certain aspects of the research question, leading to incomplete or insufficient data for analysis.
- Reliability concerns: Some research methods may suffer from reliability issues due to measurement errors or inconsistent data collection procedures.
- Learning curve: Researchers may require time and effort to master certain research methods, particularly those involving complex statistical techniques or specialized software.

Research Methodology

- Research methodology refers to the overall framework, approach, and theoretical underpinnings that guide the research process.
- It encompasses the design, planning, and execution of a research study, including the selection and integration of various research methods.
- Research methodology provides a systematic and structured approach to conducting research, ensuring that the study is valid, reliable, and meaningful.
- It involves considerations such as the research design, sampling techniques, data collection methods, data analysis procedures, and the interpretation of findings.

Research Methodology

Advantages of Research Methodology

- Provides a holistic perspective: Research methodology encompasses the entire research process, ensuring that all aspects are carefully considered and integrated.
- Enhances research quality: By following a sound research methodology, researchers can enhance the quality and rigor of their studies.
- Offers theoretical grounding: Research methodology incorporates theoretical frameworks and perspectives that guide the research process and help contextualize findings.
- Facilitates systematic data collection: Research methodology provides guidelines for data collection, ensuring that relevant and reliable data is obtained.
- Promotes transparency: A well-defined research methodology allows other researchers to understand and evaluate the methods employed, enhancing transparency.

Research Methodology

Advantages of Research Methodology

- Enables replication: Detailed research methodology enables other researchers to replicate the study, contributing to the robustness of findings.
- Supports interdisciplinary research: Research methodology can provide a common framework for interdisciplinary research, facilitating collaboration across different fields.
- Assists in data analysis: Research methodology guides researchers in selecting appropriate data analysis techniques, ensuring meaningful interpretation of results.
- Provides a foundation for research ethics: Research methodology incorporates ethical considerations, ensuring that studies are conducted in an ethical manner.
- Enables knowledge advancement: Through the use of systematic research methodology, researchers can contribute to the advancement of knowledge in their respective fields.

Research Methodology

Disadvantages of Research Methodology

- Inflexibility: Research methodology may be perceived as rigid, limiting researchers' ability to explore emerging ideas or adapt to evolving research questions.
- Time-consuming: Developing and implementing a comprehensive research methodology can be time-consuming, especially for complex or large-scale studies.
- Complexity: Research methodology can be complex, particularly when dealing with advanced statistical techniques or interdisciplinary research.
- Resource-intensive: The execution of research methodology often requires significant resources, including funding, equipment, and human capital.
- Potential for errors: Inadequate understanding or application of research methodology can lead to errors in data collection, analysis, or interpretation.

Research Methodology

Disadvantages of Research Methodology

- Lack of consensus: Different research methodologies may exist within a field, leading to debates and lack of consensus on the most appropriate approach.
- Subjectivity: Despite efforts to establish objective research methodologies, subjective judgments and biases can still influence the research process.
- Limited external validity: Research methodology may focus on internal validity, sometimes sacrificing external validity, limiting the generalizability of findings.
- Difficulty in interdisciplinary integration: Research methodology may face challenges when integrating diverse approaches or methodologies from different disciplines.
- Learning curve: Researchers may require training and familiarity with specific research methodologies, which can be time-consuming and demanding.

Research Methodology

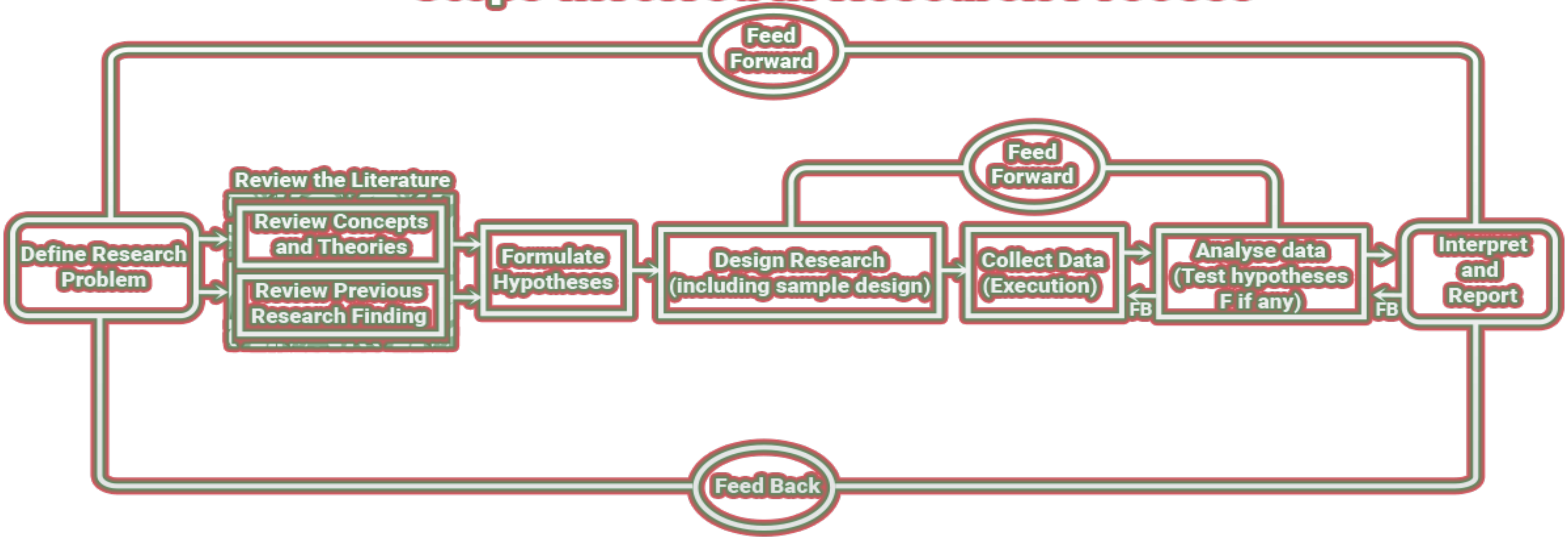
Research Method	Research Methodology
The techniques and procedures used in solving research problems	The study of research methods in other to logically justify why any particular method should be preferred to others
The major aim is to solve the research problem	The major aim is to ensure that appropriate methods are used to solve research problems
Is a component of research methodology and hence narrower in scope	Encompasses both research methods and the logic behind their adoption and hence broader in scope
Examples include surveys, experiments, interviews, observations	Examples include qualitative, quantitative, and mixed methods
Involves the use of quantitative and qualitative research methods and utilizing the knowledge and skills learned through research methodology.	Involves the learning of various techniques to conduct research and acquiring methodical knowledge to perform quantitative, qualitative, and mixed research.

Research Process

Steps Involved in Research Process:

- The research process consists of a series of actions or steps necessary to effectively carry out research and the desired sequencing of these steps.
- The research process consists of a number of closely related activities but such activities overlap continuously rather than following a strictly prescribed sequence.
- One should remember that the various steps involved in a research process are not mutually exclusive, nor they are separate and distinct.
- They do not necessarily follow each other in any specific order and the researcher has to be constantly anticipating at each step in the research process the requirements of the subsequent steps.

Steps Involved in Research Process



Research Process

Formulating the Research Problem:

- There are two types of research problems, **those which relate to states of nature** and **those which relate to relationships between variables**.
- The best way of understanding the problem is to discuss it with one's own colleagues or with those having some expertise in the matter.
- In an academic institution, the researcher can seek help from a guide who is usually an experienced man and has several research problems in mind.
- Often, the guide puts forth the problem in general terms and it is up to the researcher to narrow it down and phrase the problem in operational terms.
- In private business units or in governmental organizations', the problem is usually earmarked by the administrative agencies with whom the researcher can discuss as to how the problem originally came about and what considerations are involved in its possible solutions.

Research Process

Extensive Literature Survey:

- Once the problem is formulated, a brief summary of it should be written down.
- It is compulsory for a research worker to write a thesis for a Ph.D. degree to write a synopsis of the topic and submit it to the necessary Committee or the Research Board for approval.
- At this stage, the researcher should undertake an extensive literature survey connected with the problem.
- For this purpose, the abstracting and indexing journals and published or unpublished bibliographies are the first place to go.
- Academic journals, conference proceedings, government reports, book, etc., must be tapped depending on the nature of the problem.
- In this process, it should be remembered that one source will lead to another. A good library will be a great help to the researcher at this stage.

Research Process

Development of Working Hypotheses:

- Working hypothesis is a tentative statement made in order to draw out and test its logical or empirical consequences.
- As such the manner in which research hypotheses are developed is particularly important since they provide the focal point for research.
- They also affect the manner in which tests must be conducted in the analysis of data and indirectly the quality of data that is required for the analysis.
- The development of the working hypothesis plays an important role in maximum research.
- Hypothesis should be very specific and limited to the piece of research in hand because it has to be tested.
- The role of the hypothesis is to guide the researcher by delimiting the area of research and to keep him on the right track.
- It sharpens thinking and focuses attention on the more important facets of the problem.
- It also indicates the type of data required and the type of methods of data analysis to be used.

Research Process

Preparing the Research Design:

- The research problem having been formulated in clear cut terms, the researcher will be required to prepare a research design.
- The preparation of the research design, appropriate for a particular research problem, involves usually the consideration of the following:
    - the means of obtaining the information;
    - the availability and skills of the researcher and staff (if any);
    - explanation of the way in which selected means of obtaining information will be organized and the reasoning leading to the selection;
    - the time available for research; and
    - the cost factor relating to research, i.e., the finance available for the purpose.

Research Process

Determining Sample Design:

- The researcher must decide the way of selecting a sample of what is popularly known as the sample design.
- In other words, a sample design is a definite plan determined before any data are actually collected for obtaining a sample from a given population.

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Research Process

Collecting the Data:

- collecting the appropriate data which differs considerably in the context of money costs, time, and other resources at the disposal of the researcher.
- Primary data can be collected either through experiments or through a survey.
- If the researcher conducts an experiment, he observes some quantitative measurements, or the data, with the help of which he examines the truth contained in his hypothesis.
- It includes – observation, personal interview, telephone interviews, mailing of questionnaires, schedules, etc.
- The researcher should select one of these methods of collecting the data taking into consideration the nature of the investigation, objective and scope of the inquiry, financial resources, available time, and the desired degree of accuracy.

Research Process

Execution of the Project:

- The researcher should see that the project is executed in a systematic manner and in time.
- If the survey is to be conducted by means of structured questionnaires, data can be readily machine-processed. The survey is under statistical control so that the collected information is in accordance with the pre-defined standard of accuracy.
- If some of the respondents do not cooperate, some suitable methods should be designed to tackle this problem.

Research Process

Analysis of data:

- After the data have been collected, the researcher turns to the task of analyzing them.
- The analysis of data requires a number of closely related operations such as the establishment of categories, the application of these categories to raw data through coding, tabulation, and then drawing statistical inferences to use computers.
- Computers not only save time but also make it possible to study a large number of variables affecting a problem simultaneously.
- The researcher can analyze the collected data with the help of various statistical measures.

Research Process

Hypothesis-testing:

- The hypotheses may be tested through the use of one or more of such tests, depending upon the nature and object of the research inquiry.
- Hypothesis-testing will result in either accepting the hypothesis or in rejecting it.

Generalizations and Interpretation:

- If a hypothesis is tested and upheld several times, it may be possible for the researcher to arrive at generalization, the real value of research lies in its ability to arrive at certain generalizations.
- If the researcher had no hypothesis to start with, he might seek to explain his findings on the basis of some theory.
- It is known as interpretation. The process of interpretation may quite often trigger off new questions which in turn may lead to further researches.

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Research Process

Preparation of the Report: Finally, the researcher has to prepare the report of what has been done by him.

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Questions:

- Define research and explain its objectives, significance, and characteristics in detail.
- Discuss the importance of research in modern society with suitable examples.
- Explain different types of research in detail with examples (basic, applied, qualitative, quantitative, experimental, descriptive).
- Compare and contrast qualitative and quantitative research approaches.
- Discuss exploratory, descriptive, and experimental research in detail with examples.
- Explain the complete research process step-by-step with a diagram.

Questions:

- Differentiate between research method and research methodology with suitable examples.
- Discuss various research methods used in data collection and analysis.

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